TrES-5b

R-0670

Benchmark run · workflow W8-benchmark-deep · record `deep-bench_TRES-5_260422` · created 2026-07-27T18:29:40+00:00

BENCHMARK: NON-DETECTION

Results

Mid-Transit Time (Tmid)	2461152.8691 +/- 0.0043 BJD_TDB
Ratio of Planet to Stellar Radius (Rp/R*)	0.08 +/- 0.053
Transit depth (Rp/Rs)^2	0.64 +/- 0.84 [%]
Orbital Inclination (inc)	80.56 +/- 2.99
Airmass coefficient 1 (a1)	4.93 +/- 3.5
Airmass coefficient 2 (a2)	-1.04 +/- 0.38
Scatter in the residuals of the lightcurve fit is	59.29 %
Best Comparison Star	#2 - [540, 68]
Optimal Aperture	1.87
Optimal Annulus	3.84
Transit Duration (day)	0.03 +/- 0.028

Accuracy vs published ephemeris (ExoClock/NEA)

Rp/R■ fit vs published	0.08 ± 0.053 vs 0.142 (deviation 0.89σ)
Duration fit vs published	0.03 d vs 0.0758 d (deviation 1.25σ)
Mid-time O–C	2.9 min
Depth SNR	1.2
Residual scatter	59.29 %

Light curve

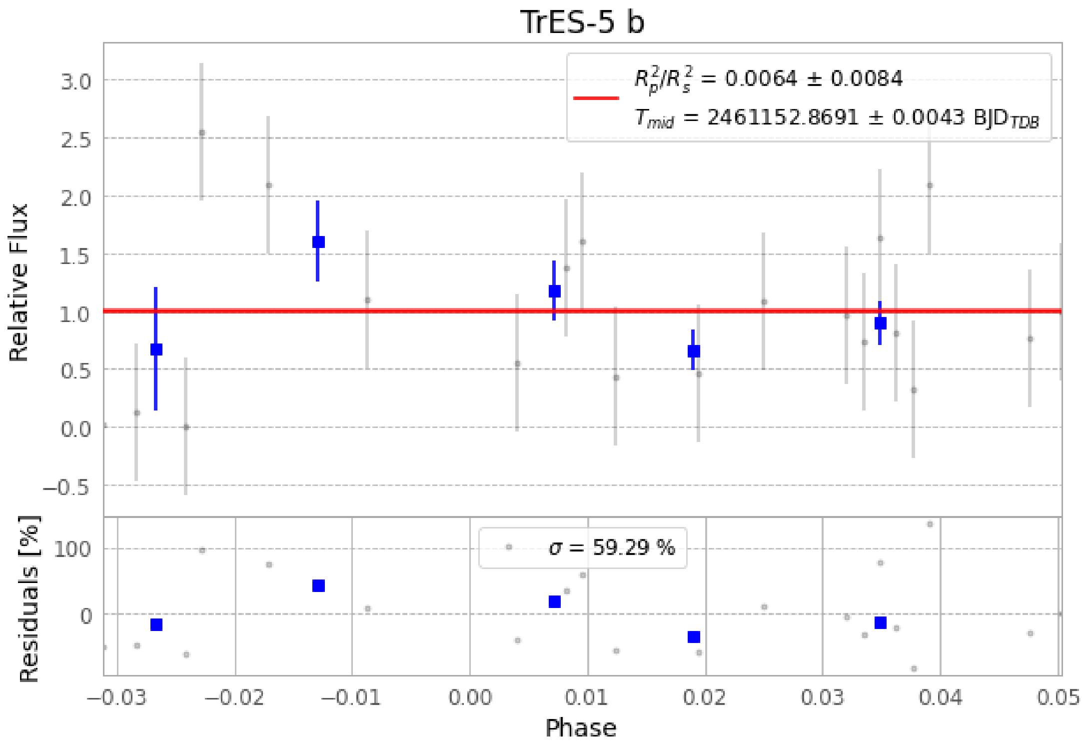

Figure 1 — FinalLightCurve_TrES-5 b_2026-04-22.png (SHA-256 35e974a3667cda80…)

Method & software

EXOTIC 4.3.1 aperture photometry with nested-sampling transit fit (ultranest 3.6.5); priors from the NASA Exoplanet Archive. Camera CCD, binning 1x1, filter CV, target pixel [480, 198], 3 comparison star(s). Exact configuration: `parameters.inits_json` in record.json (SHA-256 8f760673b48dd802…).

`exotic 4.3.1 · astropy 6.1.7 · photutils 2.0.2 · numpy 1.26.4 · scipy 1.14.1 · twirl 0.5.1 · astroquery 0.4.11 · ultranest 3.6.5 · python 3.12.13 · pipeline_commit f6a95f5`

Data provenance

61 FITS frames (40 MB), `TRES-5260422073916.FITS` → `TRES-5260422103915.FITS`. Every input frame is SHA-256-fingerprinted in record.json; raw frames available on request and verifiable against that manifest. Artifacts: bench-TRES-5-260422.log (f2618a6b788f…), FinalLightCurve_TrES-5 b_2026-04-22.png (35e974a3667c…), FinalLightCurve_TrES-5 b_2026-04-22.csv (b2aa7b1aa2ec…)

Compute resources

Wall time 255.5 s · peak memory None MB · host mac-mini-m1-8gb (macOS-26.4-arm64-arm-64bit)

Observer: — · AAVSO obscode ABGA · Automated pipeline with human review — nothing is submitted without a human approving this specific result. Generated 2026-07-27 18:29Z.